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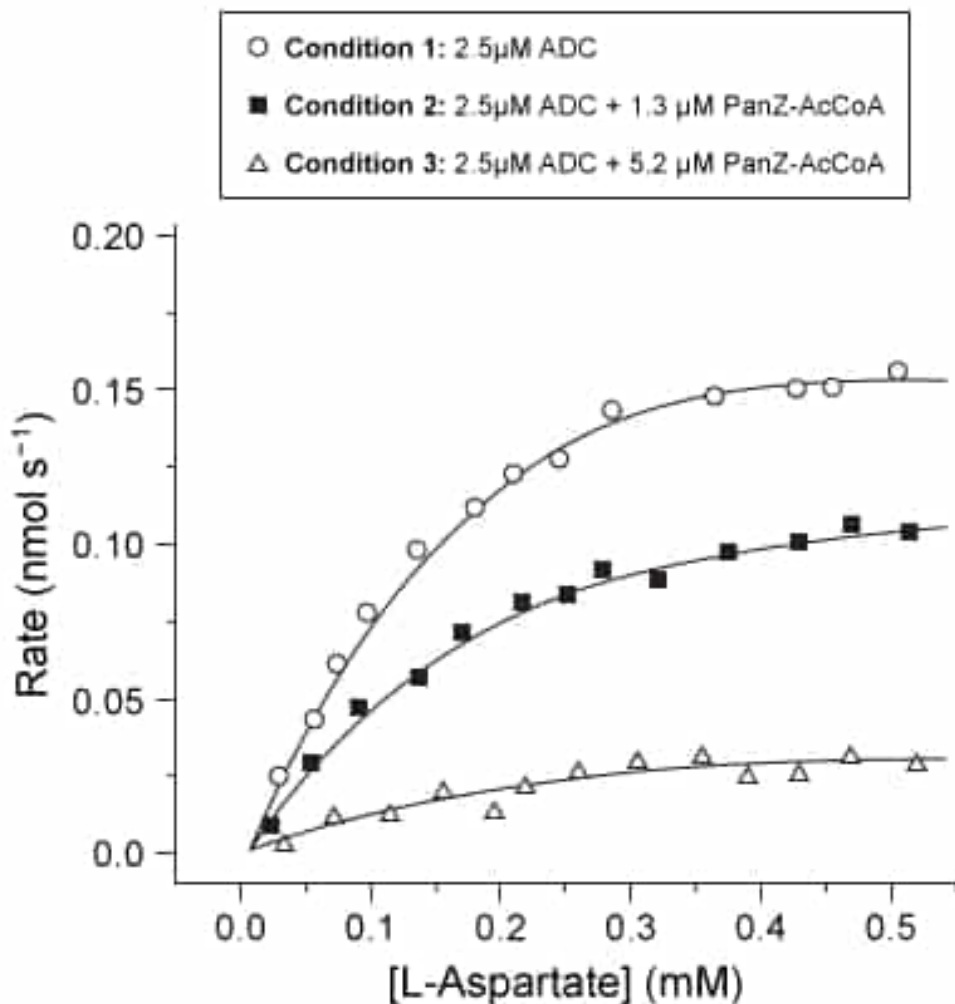


Figure 1 Catalytic activity of ADC in *E. coli* with increasing concentrations of L-aspartate

Which of the following best describes the ADC activation mechanism?

- ☐ A. AcCoA binds to the PanD-PanZ complex to induce cleavage. [15%]
- ☐ B. PanD, PanZ, and AcCoA bind in a random order to form a ternary complex. [15%]
- ☐ C. A binary complex composed of PanD and

AcCoA cleaves PanZ. [3%]

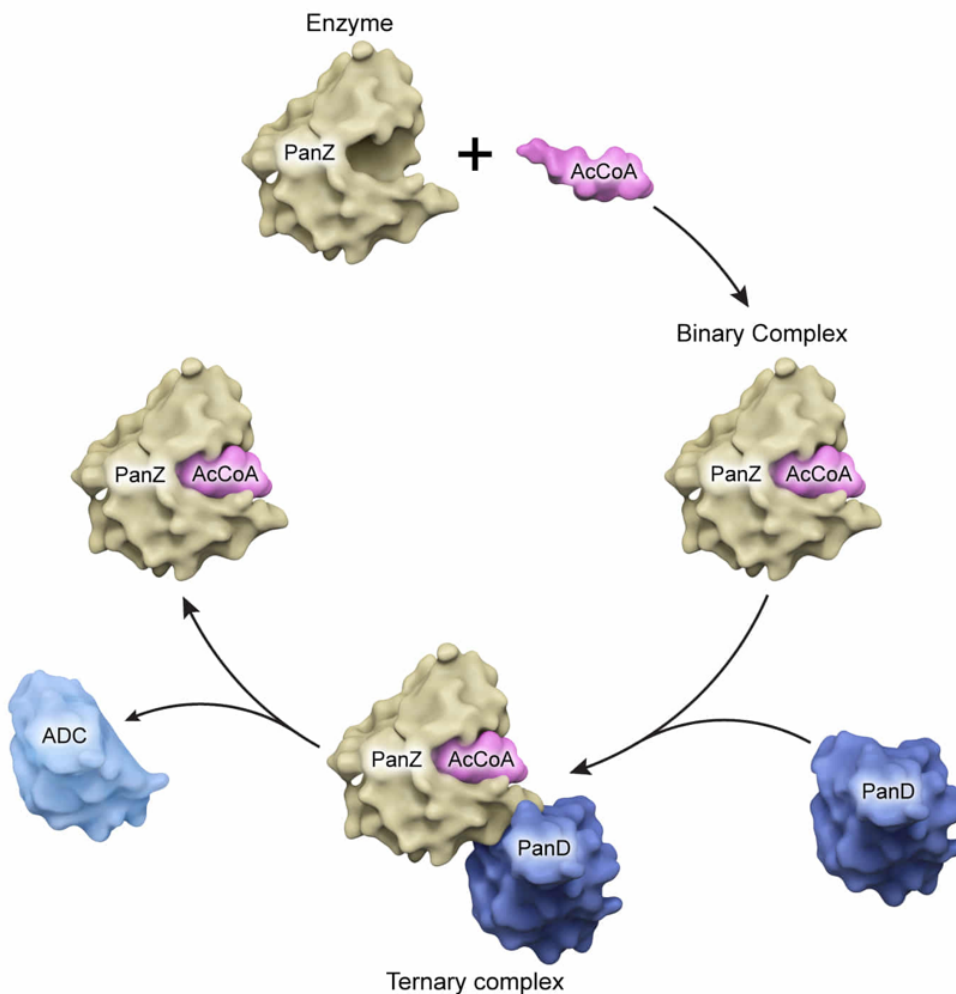
✔️ D. PanZ binds AcCoA followed by PanD to form a ternary complex. [64%]

Correct

64% answered correctly

Explanation:

ADC activation mechanism: ordered ternary complex formation



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Ternary complexes form when three distinct molecules, at least one of which is a protein, bind to each other to form a single system. They can form

through **random or ordered** mechanisms. In **random reactions**, the sequence of binding does not affect the progress of the reaction, and products are released in **no specific order**. In contrast, **ordered reactions** require that the components of the complex bind in a **defined sequence** for the reaction to proceed.

The passage states that as part of the mechanism of ADC formation, PanD binds to a complex consisting of AcCoA and PanZ. Therefore, PanZ must *already* have bound to AcCoA to form a pre-existing AcCoA-PanZ complex *before* PanD can bind. This suggests an ordered mechanism of ternary complex formation, in which PanZ binds to AcCoA and *then* to PanD.

(Choice A) AcCoA does not bind to a PanD-PanZ complex. Instead, the passage says that PanD binds to a complex of AcCoA and PanZ.

(Choice B) The PanD-PanZ-AcCoA ternary complex is not formed by a random binding mechanism. The passage says that PanD binds to an existing complex of PanZ and AcCoA, indicating that PanZ bound AcCoA *first*.

(Choice C) The binary complex that forms prior to ternary complex formation is composed of PanZ and AcCoA rather than PanD and AcCoA. PanZ-AcCoA *then* binds to PanD, inducing a conformational shift in PanD that allows it to cleave its own N-terminal region.

Educational objective:

For ternary complexes (eg, one enzyme and two substrates), an ordered reaction requires that the members be bound and released in a specific sequence. For random reactions, the order in which the members bind is not important and does not affect the progress of the reaction.

Time spent:0 sec

Subject:

Biochemistry

QID:401146

System:

Enzymes

Synthesis of coenzyme A (CoA) requires vitamin B₅ as a substrate. In bacteria, vitamin B₅ is synthesized by condensation of β -alanine and pantoic acid. β -alanine is produced by decarboxylation of L-aspartate, mediated by the enzyme aspartate α -decarboxylase (ADC). Active ADC is formed when its inactive precursor, PanD, binds to a regulatory complex composed of acetyl coenzyme A (AcCoA) and a protein known as PanZ. Formation of the PanD-PanZ-AcCoA complex causes the activation loop of PanD to fold into a β -pleated sheet, resulting in PanD autocleavage. Researchers hypothesize that this conformational change allows the Thr57 residue to cleave the N-terminal end of PanD to form ADC.

The limited information on the role of PanZ in the synthesis of CoA has stimulated an investigation of its effect on ADC activity. *Escherichia coli* cells were provided with varying concentrations of PanZ, and the rate of β -alanine production was measured in three different conditions (Figure 1). Reaction vessels were maintained at pH 6 to ensure optimal decarboxylation activity.

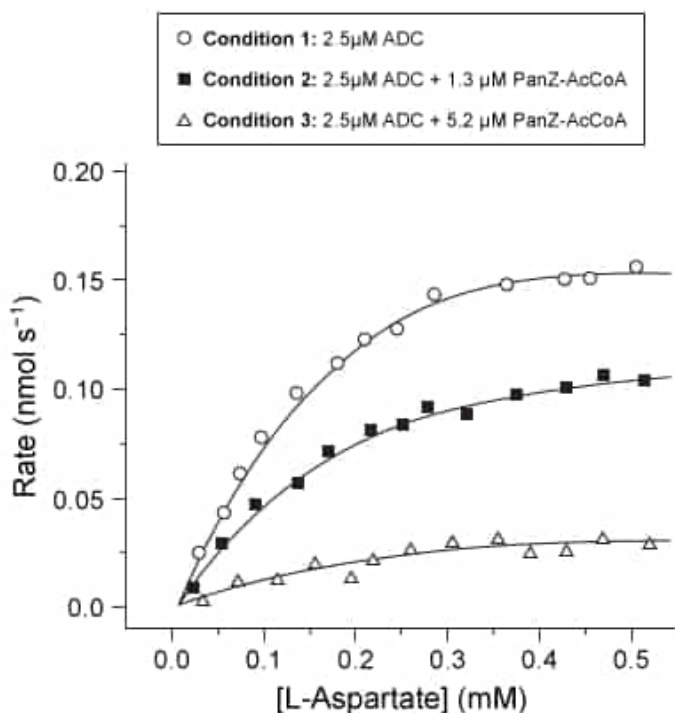


Figure 1 Catalytic activity of ADC in *E. coli* with increasing concentrations of L-aspartate

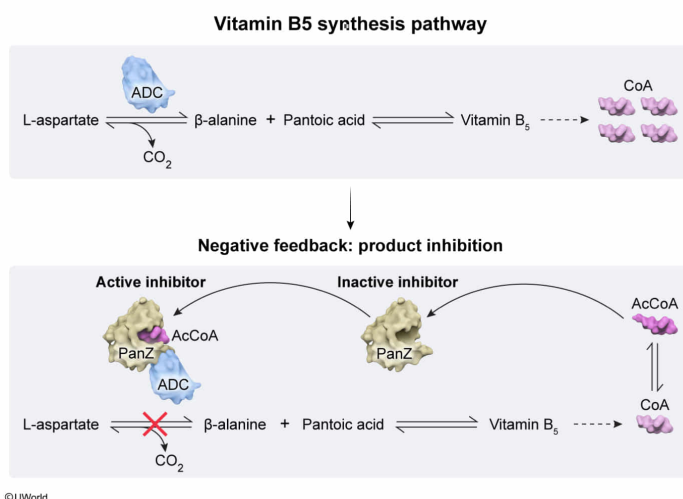
In *E. coli*, high levels of CoA lead to decreased β -alanine synthesis. This is an example of:

- ☐ A. the competitive inhibition of ADC by PanZ-AcCoA. [5%]
- ☐ B. a positive feedback loop in which CoA stimulates its own synthesis. [2%]
- ✔ ☒ C. a negative feedback mechanism driven by AcCoA involving PanZ. [81%]
- ☐ D. downregulation of PanD autocleavage, stimulated by PanZ-AcCoA. [9%]

Correct

81% answered correctly

Explanation:



Biosynthetic pathways can be regulated by negative or positive **feedback loops**. **Positive feedback** increases the amount of product synthesized by a pathway when a product activates enzymes and stimulates production of precursor molecules. **Negative feedback**, which is more common, inactivates enzymes and reduces the production of necessary precursor molecules. **Product inhibition** is a type of negative feedback mechanism in which the end product of a reaction pathway decreases its own production by binding to an enzyme or regulatory protein needed in an earlier step.

Based on the passage, β -alanine is a precursor molecule in the

production of CoA. Therefore, CoA is an end product that depends on β -alanine synthesis. The question states that *E. coli* bacteria with high levels of CoA display reduced rates of β -alanine synthesis, suggesting that accumulated CoA inhibits its own synthesis by reducing the concentration of its precursor, β -alanine. [Figure 1](#) supports this negative feedback mechanism, showing that the CoA derivative AcCoA bound to PanZ decreases the production of β -alanine by inactivating the enzyme ADC.

(Choice A) Competitive inhibitors increase the K_M value of an enzyme-substrate reaction but have no effect on V_{max} . Figure 1 shows that increasing the concentration of PanZ-AcCoA significantly decreases the V_{max} of the reaction, so PanZ-AcCoA is not a competitive inhibitor.

(Choice B) In a positive feedback loop, the accumulated CoA would *increase*, not decrease, the rate of β -alanine synthesis.

(Choice D) Based on the passage, the PanZ-AcCoA *increases* (ie, upregulates) PanD autocleavage rather than downregulating it.

Educational objective:

Positive feedback increases the amount of product synthesized by a pathway by activating enzymes and stimulating the production of precursor molecules. Negative feedback decreases the synthesis of products in a pathway by inactivating enzymes and reducing the production of necessary precursor molecules.

Time spent:0 sec

Subject:
Biochemistry

QID:401147

System:
Enzymes

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The limited information on the role of PanZ in the synthesis of CoA has stimulated an investigation of its effect on ADC activity.

Escherichia coli cells were provided with varying concentrations of PanZ, and the rate of β -alanine production was measured in three different conditions (Figure 1). Reaction vessels were maintained at pH 6 to ensure optimal decarboxylation activity.

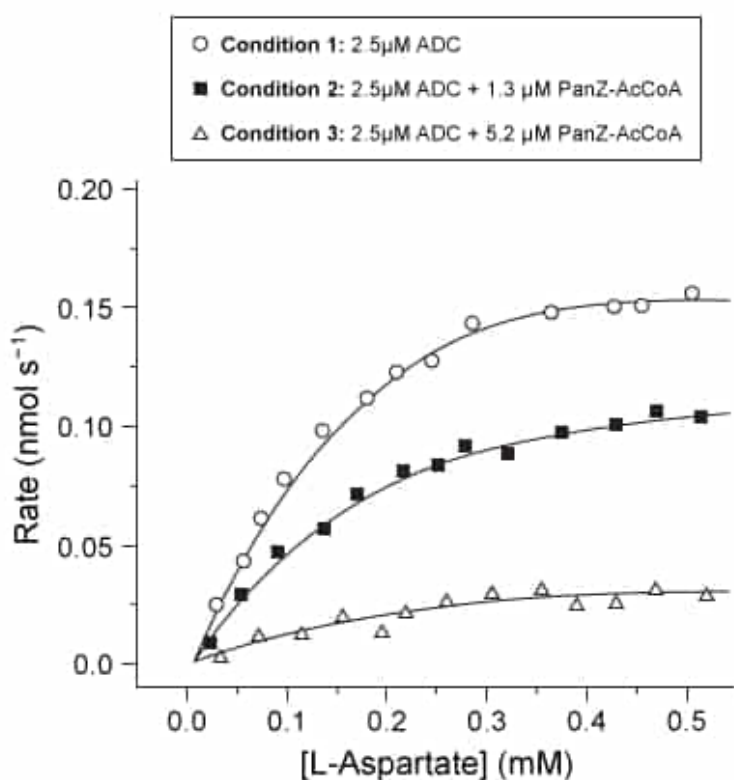


Figure 1 Catalytic activity of ADC in *E. coli* with increasing concentrations of L-aspartate

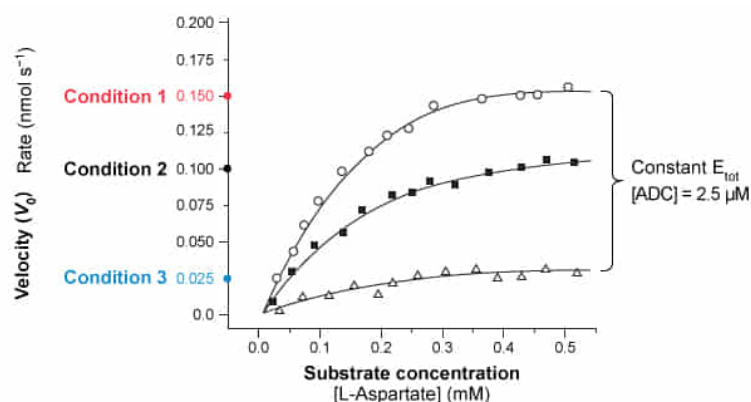
Based on the results shown in Figure 1, what is the ratio of k_{cat} in Condition 1 to k_{cat} in Condition 3 for ADC?

- ✓ ☒ A. 6:1 [73%]
☐ B. 4:1 [13%]
☐ C. 1:2 [3%]
☐ D. 1:6 [8%]

Correct

73% answered correctly

Explanation:



1. **Condition 1** **Condition 3**

$$k_{\text{cat}1} = \frac{V_{\text{max}1}}{E_{\text{tot}}} \quad k_{\text{cat}3} = \frac{V_{\text{max}3}}{E_{\text{tot}}}$$

2. **Ratio**

$$\frac{k_{\text{cat}1}}{k_{\text{cat}3}} = \frac{V_{\text{max}1}}{E_{\text{tot}}} \times \frac{E_{\text{tot}}}{V_{\text{max}3}}$$

$$= \frac{V_{\text{max}1}}{V_{\text{max}3}} = \frac{0.150 \text{ nmol s}^{-1}}{0.025 \text{ nmol s}^{-1}}$$

$$= \frac{6}{1} \text{ or } \boxed{6:1}$$

Figure 1 is a Michaelis-Menten graph that plots the rate of an

enzyme-catalyzed reaction V_0 against the substrate concentration $[S]$. The rate of the reaction V_0 gradually increases as $[S]$ increases and finally reaches a **maximum velocity** $V_{\max}$, at which point the enzyme's active sites are **saturated** with substrate. In this state, the **turnover number** k_{cat} , with units of reciprocal seconds (s^{-1}), represents the maximum number of substrates converted to product by a single enzyme each second. The maximum combined rate of all the enzymes in a solution $V_{\max}$ is directly proportional to k_{cat} . As a result, k_{cat} can be determined by dividing $V_{\max}$ by the **total enzyme amount** E_{tot} .

The differences in the enzymatic activities of ADC in the presence (Condition 2 and 3) or absence (Condition 1) of PanZ-AcCoA are shown in Figure 1. The question asks for the ratio of the k_{cat} values of Condition 1 to those of Condition 3. The total enzyme concentration $[E_{\text{tot}}]$ is $2.5 \mu\text{M}$ in each condition, so the k_{cat} value for each reaction is proportional to $V_{\max}$. Therefore, the k_{cat} ratio of Condition 1 to Condition 3 is equal to the ratio of their respective $V_{\max}$ values. Quantitatively, this is represented as 0.15

$\text{nmol} \cdot \text{s}^{-1} / 0.025 \text{ nmol} \cdot \text{s}^{-1}$, which can be simplified to a 6:1 ratio.

(Choice B) This is the k_{cat} ratio of Condition 2 to Condition 3.

(Choice C) This k_{cat} ratio represents a close approximation of the comparison of Condition 1 to Condition 2.

(Choice D) This k_{cat} ratio is the reverse of Condition 1 to Condition 3.

Educational objective:

The turnover number k_{cat} is the maximum number of substrates

converted to products by a single enzyme molecule each second. The value k_{cat} is proportional to the maximum velocity V_{max} of the reaction and is calculated as $V_{\text{max}}/[E_{\text{tot}}]$ (total enzyme concentration).

Time spent:0 sec

Subject:
Biochemistry

QID:401148

System:
Enzymes

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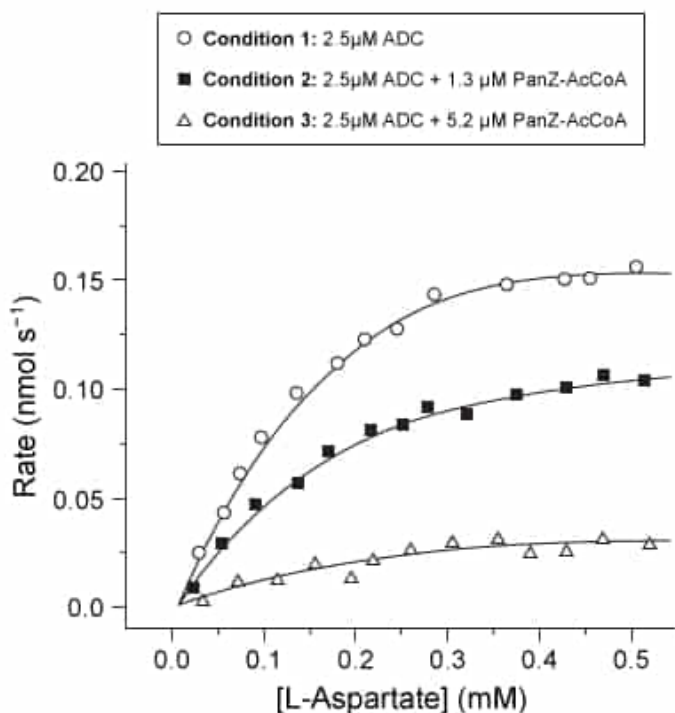


Figure 1 Catalytic activity of ADC in *E. coli* with increasing concentrations of L-aspartate

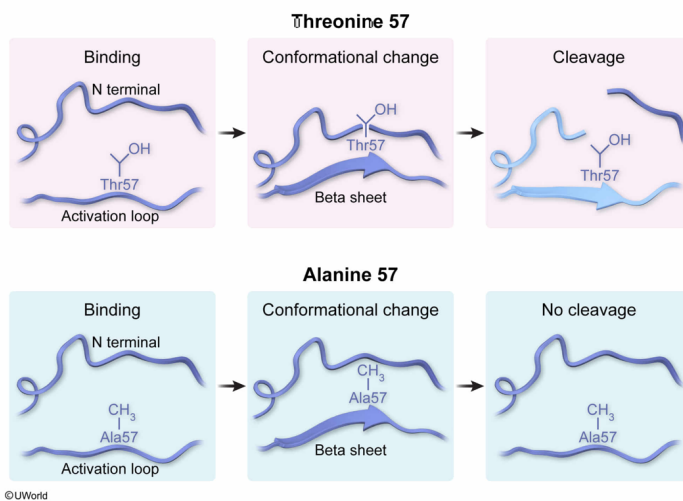
Which experiment could best provide data to support the proposed mechanism of the autocleavage reaction described in the passage?

- ☐ A. Evaluate PanZ's structure with circular dichroism in the presence and absence of AcCoA. [9%]
- ✔ ☒ B. Measure the rate of ADC formation after site-directed mutagenesis of Thr57 to alanine in PanD. [70%]
- ☐ C. Measure the molecular weight of PanD in the absence of PanZ and AcCoA. [16%]
- ☐ D. Calculate the molecular weight of ADC and isolate it through size-exclusion chromatography. [3%]

Correct

69% answered correctly

Explanation:



Enzymatic activity generally requires a **specific set of amino acid residues** in the active site of the enzyme to catalyze the reaction of interest. **Replacing** any of these critical amino acids with another amino acid typically **results in inactivation** of the enzyme. Therefore, the role of a specific amino acid in enzyme activity can be determined by deliberately substituting it for a different amino acid.

The researchers hypothesized that the PanD autocleavage mechanism involves a conformational change in PanD that allows Thr57 to cleave the N-terminal portion of the protein from the rest of the chain. If this is

correct, mutating Thr57 to alanine most likely eliminates enzymatic activity whereas if Thr57 is not involved in autocleavage, mutating it most likely has little effect. Although additional experiments are needed to verify the role of Thr57 in autocleavage, the abolishment of autocatalytic activity upon mutation of Thr57 **provides evidence to support** its role in the catalytic mechanism described in the passage. In contrast, the other available choices do not provide any information about the mechanism of autocleavage.

(Choice A) Circular dichroism can assess secondary structure (α -helices and β -sheets) in proteins. The passage says that PanD, not PanZ, must form a new β -pleated sheet to facilitate autocleavage. Therefore, the structure of PanD, not PanZ, would need to be probed to provide evidence of the proposed mechanism.

(Choice C) PanZ-AcCoA is required in the autocleavage mechanism.

Therefore, the molecular weight of PanD will not change if PanZ and AcCoA are absent because autocleavage will not occur.

(Choice D) [Size-exclusion chromatography](#) separates proteins by size and could be used to distinguish ADC from PanD, but it would not describe the mechanism of autocleavage from PanD to ADC.

Educational objective:

Enzymatic activity is dependent on the correct sequence and conformation of specific amino acids in the enzyme active site. Mutating critical amino acids generally inhibits enzymatic function. Deliberately mutating amino acids of interest can help reveal their role in enzymatic function.

Time spent:0 sec

Subject:
Biochemistry

QID:401149

System:
Enzymes

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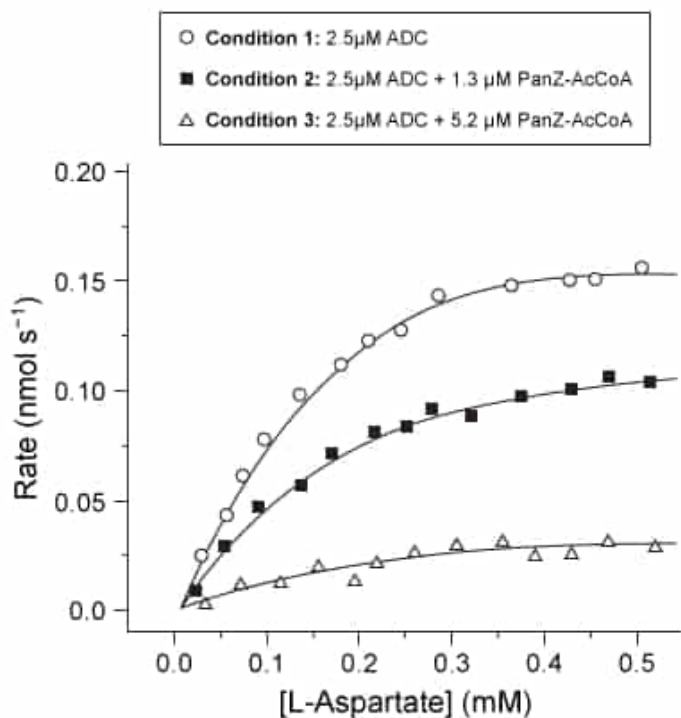
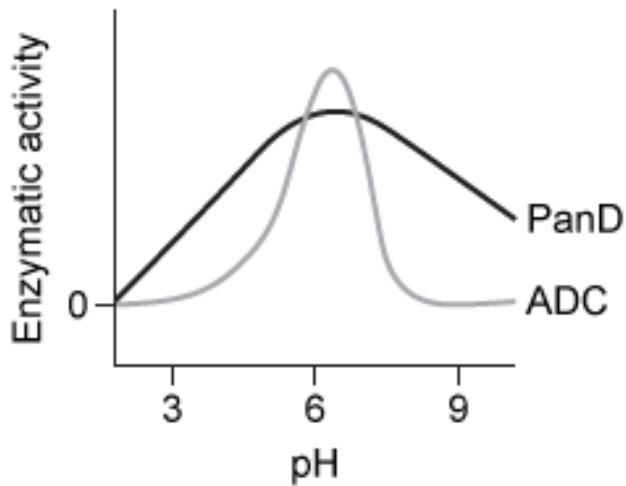


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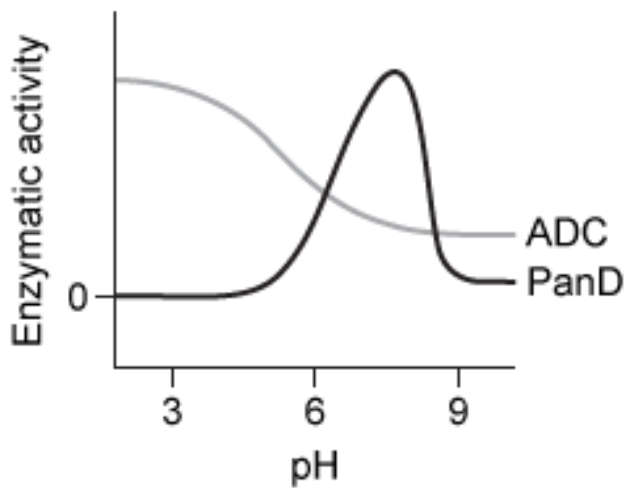
Which of the following graphs accurately predicts the aspartate decarboxylation activity of PanD and ADC between pH 3 and pH 9?

☐ A.



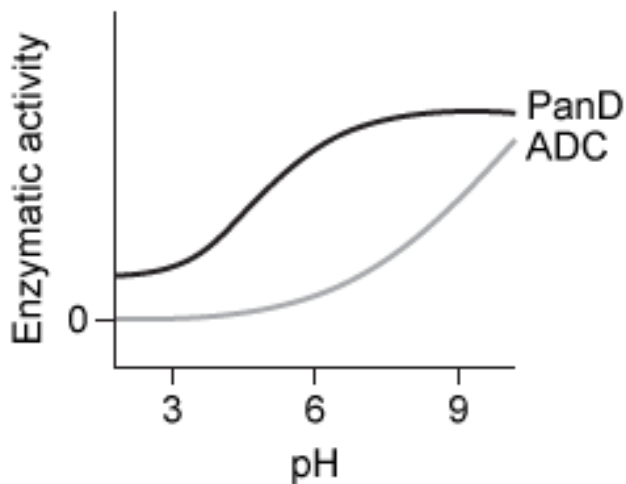
[46%]

☐ B.



[9%]

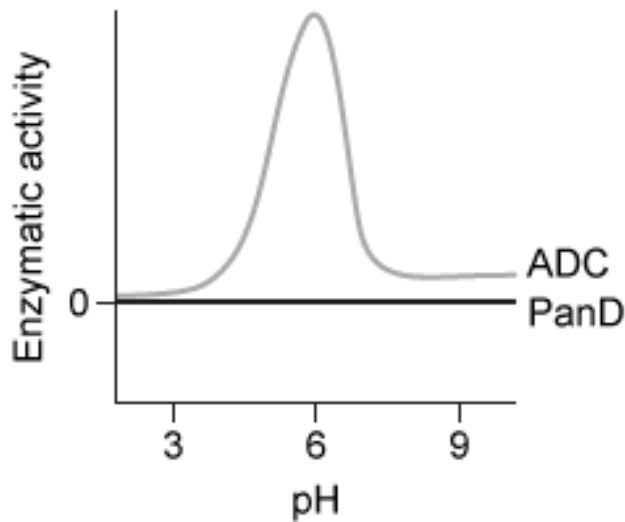
☐ C.



150/1

[37%]

✓ ☒ D.

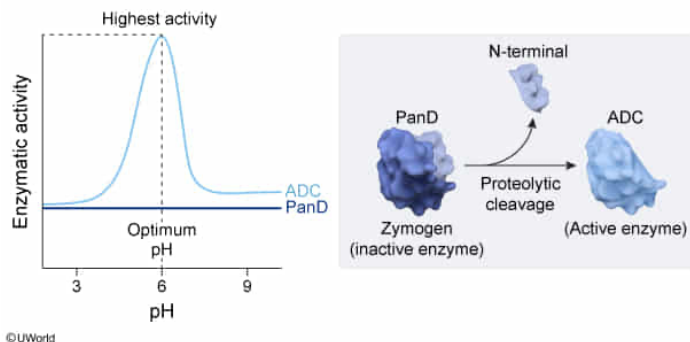


[38%]

Correct

38% answered correctly

Explanation:



Changes in the pH of the aqueous environment can alter the charge and folding of proteins. Enzymes in particular require a specific **pH range** to maintain their activity and structure. The highest enzymatic activity is achieved at a particular pH that is specific to the enzyme (ie, **optimal pH**). Activity is greatly reduced at pH values that are significantly lower or higher than the optimal value.

The passage states that reactions were maintained at pH 6 to ensure that ADC reached its highest enzymatic activity. Therefore, a graph of ADC activity ranging from pH 3 to pH 9 would appear as a bell curve with a peak at pH 6. All other pH levels should show lower activity.

The passage also states that **PanD is the inactive form** of the enzyme ADC and cannot decarboxylate L-aspartate. Enzymes in their inactive

forms, referred to as **zymogens or proenzymes**, must be altered by covalent modifications to become active. In the case of PanD, **proteolytic cleavage** (proteolysis) is used to activate the proenzyme. Because PanD is inactive, it displays **no aspartate decarboxylation activity at any pH**. Only the graph shown in Choice D illustrates both the correct optimal pH of ADC and the inactivity of PanD.

(Choices A, B, and C) These graphs incorrectly show aspartate decarboxylation activity for PanD. Graphs B and C also incorrectly show optimal ADC activity at a pH other than 6.

Educational objective:

The activity of an enzyme is greatest at the optimal pH of that enzyme, and gradually decreases at pH values that are significantly lower or higher. Proenzymes or zymogens are the inactive forms of enzymes that require post-translational modifications to become activated. Proteolytic cleavage (proteolysis) is one such post-translational modification that involves the breakdown of proteins into shorter polypeptides or amino acids by enzymes known as proteases.

Time spent:0 sec

Subject:
Biochemistry

QID:401150

System:
Enzymes

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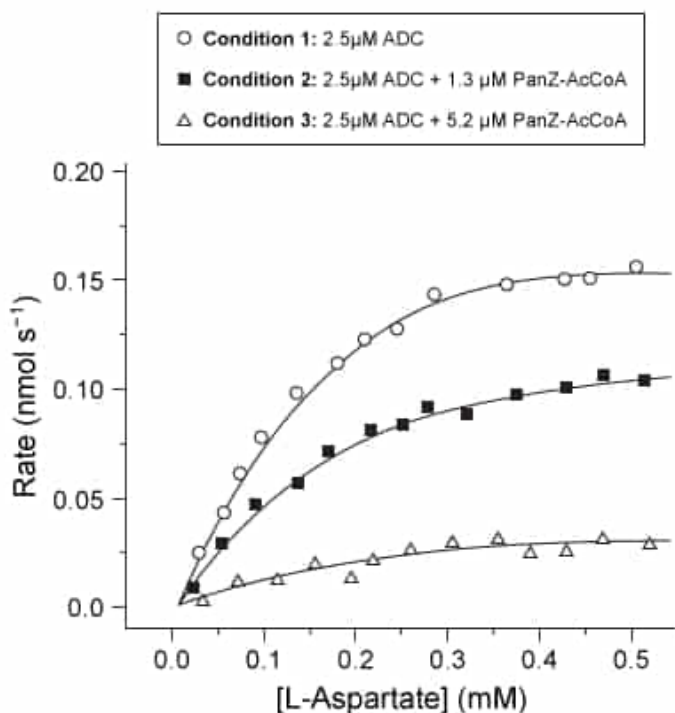


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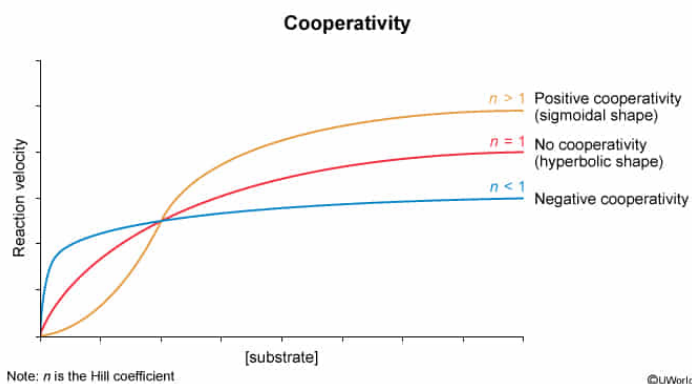
ADC isolated from the Gram-negative bacteria *Alcaligenes faecalis* showed a sigmoidal dependence on L-aspartate concentration. Which of the following could be the Hill coefficient for this form of ADC?

- ☐ A. 0 [1%]
- ☐ B. 0.5 [12%]
- ☐ C. 1 [22%]
- ✓ ☒ D. 2 [63%]

Correct

63% answered correctly

Explanation:



An enzyme or protein with multiple subunits is described as cooperative if a ligand or substrate binding to one subunit changes the binding affinity of other subunits. The **Hill coefficient n** is a quantitative measure of cooperativity obtained through kinetic analysis. It can be determined by fitting the data to a modified form of the Michaelis-Menten equation:

$$V_0 = \frac{V_{\max}[S]^n}{K_M^n + [S]^n}$$

Enzymes that exhibit *increased* binding affinity when the first subunit binds its substrate are described as positively cooperative ($n > 1$); these enzymes display sigmoidal (s-shaped) dependence on substrate concentration. In contrast, some enzymes show *decreased* binding affinity after the first subunit binds its substrate. These enzymes are said to be negatively cooperative ($n < 1$) and show hyperbolic dependence on

substrate concentration. In addition, many enzymes exhibit no cooperativity ($n = 1$) and display hyperbolic dependence on substrate concentration.

The ADC obtained from *Alcaligenes faecalis* displays **sigmoidal dependence** on the concentration of its substrate, L-aspartate. Therefore, it exhibits *positive* cooperativity and must have a **Hill coefficient greater than 1**.

(Choice A) A Hill coefficient of 0 indicates infinitely negative cooperativity. All enzymes have a Hill coefficient greater than 0.

(Choice B) A Hill coefficient of 0.5 indicates negative cooperativity and would show hyperbolic (*not* sigmoidal) dependence on substrate concentration.

(Choice C) A Hill coefficient of 1 indicates no cooperativity and would appear as a hyperbolic curve rather than a sigmoidal curve.

Educational objective:

The Hill coefficient n is a quantitative measure of cooperativity. Enzymes in which $n > 1$ display positive cooperativity and have sigmoidal dependence on substrate concentration. Enzymes where $n = 1$ or $n < 1$ exhibit no cooperativity or negative cooperativity, respectively, and exhibit hyperbolic kinetics.

Time spent:0 sec

Subject:
Biochemistry

QID:401151

System:
Enzymes